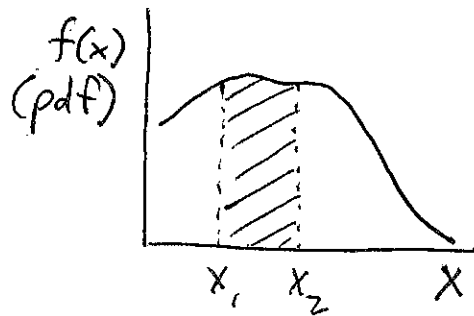


⇒ Previously

- IIDRV (Independent, Identically Distributed Random Variable)
- Probability Density



$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} f(x) dx$$

- Descriptive Statistics: Mean, Std Dev, Skewness, Kurtosis
"Moments" of Distribution

Mean $\bar{X} = \frac{1}{N} \sum_{i=1}^N x_i$ $\sigma_{\bar{X}} = \frac{S}{\sqrt{N}}$

STD DEV $S = \sqrt{\frac{1}{(N-1)} \sum_{i=1}^N (x_i - \bar{X})^2}$ $\sigma_S = \frac{S}{\sqrt{2(N-1)}}$

- Lab 4: Exponential Distribution

$$f(x) = \frac{1}{\beta} e^{-x/\beta}$$

$$\bar{X} = \sigma = \beta$$

- Today: Continue investigation of
Method of Moments

$\Rightarrow$ Central Limit Theorem (HW02)

Given an arbitrary distribution $f(x)$, characterized by mean μ and std dev σ , the mean of N values $\{x_i\}$ drawn from that distribution will follow a Gaussian

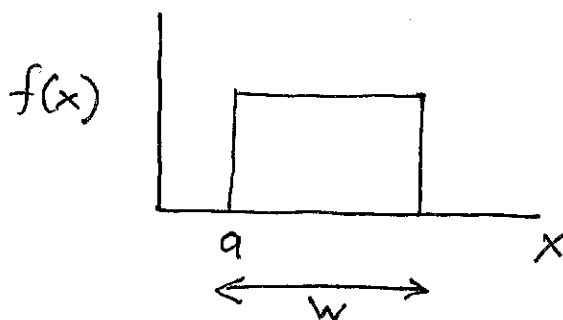
$$\mathcal{N}(\mu, \frac{\sigma}{\sqrt{N}})$$

with approximation improving as $N \rightarrow \infty$

{ n.b. NOT QUITE ARBITRARY... THIS MUST FALL OFF FASTER THAN $1/x^2$ FOR $x \rightarrow \infty$. Cauchy Distribution IS A COUNTEREXAMPLE. }

< Notebook: Central Limit Theorem >

Uniform Distribution, width W



$$\bar{x} = a + W/2$$

$$\sigma_x = \frac{W}{\sqrt{12}} = 0.2887 W$$